

One Input Representation for Five Chemogenomic Datasets

What a shared genotype channel carries, why the dose axis does not pool, how far the compound panels reach, and whether the metabolites can join

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Abstract

Five served datasets hold 7.48 million chemogenomic records, and training one model across them requires a single input representation. They disagree on three axes. Ploidy runs from haploid through homozygous diploid to heterozygous diploid. Compound panels run from 41 to 5,170. Dose runs from an unquantified target inhibition through a nine-order-of-magnitude molar range to a single fixed concentration. Every number here is measured from the served records through their loaders.

The genotype axis resolves cleanly. The schema already carries heterozygosity in two parts, a genome-wide ploidy baseline and a per-locus copy-number deviation, and all five datasets round-trip through it. A per-gene functional dose, copies present over copies in the reference, is lossless for these five because every perturbation is either a full deletion or a one-of-two variant, and it needs no separate ploidy feature because the vector’s background level is the ploidy. It is also learnable rather than merely expressible: Hoepfner supplies 648,977 gene-by-compound cells measured both heterozygous and homozygous. The gap is entirely in the tensor layer, which reads only a gene name and therefore makes those 648,977 pairs byte-identical inputs carrying different responses.

The dose axis does not resolve. Forty of Vanacloig’s 41 compounds carry no concentration at all, only a target-inhibition basis; every Wildenhain compound sits at one fixed concentration, so its dose column is constant; and for the 214 compounds dosed in more than one dataset the median ratio between the highest and lowest dose is 12.5. Dose therefore enters as a basis token rather than a number.

Chemically, pooling buys coverage of the phenolic inhibitors and not of the small alcohols, and Wildenhain supplies essentially all of the gain. Yeast9 can share the molecule space for the two thirds of its species that have a structure, and 103 of those are dosed somewhere in the five datasets.

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Contents

1	The question 	2
2	The five datasets 	2
3	The genotype axis resolves 	2
4	The dose axis does not resolve 	3
5	What pooling buys chemically 	4
6	Whether the metabolites can join 	4
7	The representation 	5

1 The question x

Five chemogenomic datasets are served and together hold 7.48 million records. Training one model on all of them is attractive for an obvious reason and blocked by a less obvious one. The attraction is scale and chemical breadth: no single dataset has both a genome-wide gene panel and a large compound panel. The block is that a model takes one input format, and the five datasets disagree about what a strain is and about what a dose is.

Three axes carry the disagreement. Ploidy runs from haploid single deletions through homozygous diploids to heterozygous diploids where one wild-type copy survives. Compound panels run from 41 to 5,170. Dose runs from an unquantified target inhibition, through a molar range spanning nine orders of magnitude, to a single fixed concentration applied to every compound.

The application target is Vanacloig-Pedros 2022 (10.1128/spectrum.02017-22), the only served screen that doses the industrial hydrolysate panel at a matched target inhibition and the only one in which isobutanol appears as a stress. The others enter as partners: Hillenmeyer 2008 (10.1126/science.1150021) and Hoepfner 2014 (10.1016/j.micres.2013.11.004) for record count and gene coverage, Wildenhain 2015 (10.1016/j.cels.2015.12.003) for compound diversity.

This document measures what a shared representation can absorb and what it cannot, and ends with the representation itself in Section 7.

2 The five datasets x

Table 1 puts the five on the axes that matter for a shared input. Record counts and gene counts are read from the flattened served records; compound counts are distinct InChIKeys with a dosed record; the dose columns are the stored basis and the share of records whose unit converts to molar without a molecular weight.

Table 1: The five served chemogenomic datasets. Records and genes come from the flattened served records, compounds are distinct InChIKeys with a dosed record, and the last column is the share of records whose dose unit converts to molar.

dataset	records	genes	compounds	ploidy	dose basis	molar
Vanacloig 2022	143,218	3,598	41	haploid	IC30, fixed	0%
Hillenmeyer HOM	1,088,620	4,675	114	diploid	fixed	84%
Hillenmeyer HET	2,698,797	5,825	290	diploid	fixed	96%
Hoepfner 2014	3,124,319	5,842	148	diploid	IC30	100%
Wildenhain 2015	428,206	242	5,170	haploid	none	100%

Three observations set up the rest of the document. Hoepfner is the largest store and the only one holding both ploidy arms, 1,759,255 heterozygous records and 1,365,064 homozygous, over a shared gene universe. Wildenhain has eighteen times the compound panel of the next largest and a fifth of the gene panel. Vanacloig is the smallest and is the application target, and it is the only dataset whose dose is not a number.

One correction worth recording, because it was wrong in an earlier reading of the raw files. The Hoepfner score matrices are genes by screens, not screens by genes: the roughly 5,900 columns per file are 2,956 experiments run in duplicate, and the 6,681 rows are the genes.

3 The genotype axis resolves x

A homozygous diploid deletion can be written as the haploid genotype plus a statement of ploidy, because both copies are gone. A heterozygous deletion cannot, because one copy survives and the gene sits at half dose. The question is whether the schema can express that and whether the modeling stack can consume it. The two answers differ.

The schema already carries it. Heterozygosity is stored in two parts. A genome-wide baseline sits on the reference genome’s ploidy field, which is the copy number of every unperturbed gene, and a per-locus deviation sits on an engineered copy-number perturbation carrying the copies present and the copies in the reference. That extension was made for exactly this case. Read from the built stores, all five datasets round-trip: Vanacloig and Wildenhain are haploid with deletion leaves, the Hillenmeyer homozygous arm and the Hoepfner homozygous arm are diploid with deletion leaves, and the Hillenmeyer heterozygous arm and the Hoepfner heterozygous arm are diploid with a one-of-two copy-number leaf marked present. Heterozygous and homozygous records are distinguishable from the stored record alone on three redundant signals: the perturbation class, the state field, and whether the copy-number fields exist.

The tensor layer discards it, and that is a correctness problem. The processor that turns a genotype into model input reads exactly one field off every perturbation, the systematic gene name. Neither copy number nor ploidy appears anywhere in the dataset-to-tensor layer. A heterozygous record and a homozygous record for the same gene therefore produce byte-identical input, one index and one boolean each.

That matters quantitatively because Hoepfner measures both. Its two arms share 4,885 genes and 147 compounds, and **648,977 gene-by-compound cells are measured both heterozygous and homozygous**. Under the current representation those 648,977 pairs are identical inputs carrying two different measured responses, which is label noise rather than a missed feature.

One channel closes it. A per-gene functional dose, copies present over copies in the reference, is lossless for these five datasets on the dosage axis, because every perturbation in all five is either a full deletion or a one-of-two engineered variant. It needs no separate ploidy feature, because the vector's background level is the ploidy: a haploid deletion sits at zero against a background of one and a homozygous diploid deletion sits at zero against a background of two, so the two remain distinguishable. It is a strict generalization of the present boolean, since every current dataset emits zero for a deletion and reproduces current behavior exactly.

Four things the channel does not carry, all measured rather than supposed. Cassette identity is lost, and this is not a detail: a KanMX replacement inserts a heterologous gene that is not in the S288C universe, so no vector over yeast genes can hold it. Barcode and collection are lost, and they are the readout identity in a pooled assay. Allele classes off the dosage axis are lost, which is the hard limit for joining this work to the trigenic datasets, since a temperature-sensitive or DAmP allele sits at full copy number and is functionally compromised. And perturbation cardinality becomes a strain property rather than a constant, since Vanacloig contributes four indices per record against one for the others.

A flagged risk for the metabolic module. The flux layer hard-zeros a deleted gene's slot in the genome-scale model. A heterozygous deletion is not a zero there, so admitting heterozygous data without changing that step asserts a full knockout of a gene that still has a working copy.

4 The dose axis does not resolve x

A pooled model sees one dose field, and a field that means different things in different rows is a nuisance variable rather than a feature. The five datasets use three mutually non-convertible regimes, and the measurement is worse than a missing column.

Vanacloig's dose is a potency, not a quantity. Of its 143,218 records, 3,492 carry a numeric concentration and they are all one compound, benomyl. The other 40 of 41 compounds carry a target-inhibition basis with the concentration unstated. There is no molar value to pool.

Wildenhain's dose is a constant. Every one of its 5,170 compounds sits at the same fixed concentration, so its dose column has zero variance and carries no information at all. This is harmless for a model that ignores dose and fatal for one that regresses on it.

Hillenmeyer is the only arm with real variation. Its two arms carry a numeric dose on essentially every record, converting to molar for 84 and 96 percent of them, across a range spanning nine orders of magnitude, with up to seven doses per compound.

Hoepfner is the only possible bridge, and it is two compounds wide. It is the only dataset stating both a molar concentration and a target-inhibition basis, on 100 percent of its records, with a median of two doses per compound. In principle that calibrates what the basis means in molar terms, which is exactly what Vanacloig lacks. In practice Vanacloig and Hoepfner share two compounds, so the calibration cannot reach the panel.

Shared compounds are not dosed alike. Across the five datasets, 214 compounds are dosed in more than one with a molar value. The median ratio between the highest and the lowest of those doses is 12.5, and the 90th percentile is 100. A compound shared by two datasets is typically dosed an order of magnitude apart in them.

What follows. Dose cannot be one shared continuous feature. Three routes exist and they are not equivalent. A dose-basis token per dataset alongside log molar where it exists keeps the information and accepts that the model cannot compare across bases. Dropping absolute dose and training on within-condition gene profiles removes the problem and makes Wildenhain's constant harmless. Learning a per-compound potency scale from the multi-dose compounds and using it to place the single-dose ones is a modeling project rather than a preprocessing step. The representation in Section 7 takes the first, because it is the only one that keeps the information without a new model.

5 What pooling buys chemically ✗

Pooling is worth the trouble only if the partner panels put chemistry near the application panel. Table 2 measures that for every Vanacloig compound: its most similar compound in each partner, under Tanimoto on count ECFP4. The fingerprint metric is the one reported because cosine between dense embeddings is high for any two molecules and says nothing about closeness.

Table 2: Coverage of the 41-compound Vanacloig panel by each partner. Exact counts identical InChIKeys; the median nearest neighbor and the threshold counts use non-exact neighbors only, so an exact match cannot inflate them.

source	exact	median NN	above 0.5	above 0.7
union of others	12	0.600	13	3
Wildenhain 2015	10	0.545	13	3
Hillenmeyer HOM	2	0.327	1	0
Hillenmeyer HET	2	0.317	9	1
Hoepfner 2014	2	0.305	0	0

Wildenhain supplies essentially all of the gain. Exact overlap with the panel rises from two compounds against either Hillenmeyer arm to ten against Wildenhain and twelve against the union, and the median nearest neighbor rises from 0.33 and 0.32 to 0.60. Hoepfner contributes nothing chemically: two exact matches, a median nearest neighbor of 0.29, and no non-exact neighbor above 0.5.

The gain is uneven, and it fails where the application needs it. Sorting the hydrolysate panel by its nearest neighbor in the union of the other four, ferulic acid becomes an exact match, vanillin reaches 0.62 against it, syringaldehyde 0.60 against syringic acid, and coumaric acid 0.55 against caffeic acid. The phenolics are genuinely covered. Isobutanol improves from 0.14 against sorbitol, which was its best neighbor when only Hillenmeyer was available, to 0.47, but its neighbor is an amino alcohol rather than an alcohol. Furfural sits at 0.30 against benzaldehyde and ethanol at 0.31. At two and five heavy atoms the small alcohols are nearly empty under a substructure fingerprint, so this is a limit of the representation as much as of the panel.

Pooling therefore buys phenolic coverage and does not buy alcohol coverage, and isobutanol is the application target.

6 Whether the metabolites can join ✗

The plan is to represent media components and metabolic-model metabolites as molecules in the same embedding space as the dosed inhibitors. Two things decide whether that works: how many Yeast9 species have a structure at all, and whether the encoders can embed them.

The encoders are not the constraint. Eleven of the twelve registered encoders embed every structure-bearing species with no failures (Table 3). The three-dimensional encoder reaches 836 of 894, its refusals being monatomic ions with no conformer and planar or linear molecules, together with the inositol-phosphoceramide sphingolipids. The descriptor block emits undefined entries on 18 rows, so a caller imputes before taking any distance.

Table 3: Encoder coverage over the 894 structure-bearing Yeast9 species.

encoder	dim	n_attempted	n_embedded	n_failed
ecfp4_count	2048	894	894	0
ecfp4_bit	2048	894	894	0
fcp4_count	2048	894	894	0
maccs	167	894	894	0
rdkit_2d	217	894	894	0
mol2vec	300	894	894	0
chemberta2_mlm	384	894	894	0
chemberta2_mtr	384	894	894	0
molformer_xl	768	894	894	0
roberta_zinc_480m	768	894	894	0
unimol_v1	512	894	836	58
mole_static	1000	894	894	0

The structures are the constraint. Yeast9 holds 2,806 compartment-specific entries that collapse to 1,378 distinct species, and only 894 of those carry a SMILES, all from the release’s own name-keyed table. No species carries an InChI, so every key used here is derived from the shipped SMILES rather than sourced. Two quality problems matter for a shared space. Every shipped SMILES is stereochemically flat, with 621 of 894 carrying an unassigned stereocenter, so

L and D forms collapse onto one key. And 36 do not encode the species the model declares, judged against the model’s own formula, two of them being outright wrong molecules.

The 484 species without a structure split into two halves that need different treatment (Table 4). About 167 have no single structure and never will, being transfer RNAs, pooled lipid classes, protein species, glycan accessions and biomass pseudo-metabolites. Those need a non-molecular slot regardless of curation. The remaining 317 have a definite structure absent from the release table, dominated by acyl-resolved lipids, and 295 of the 484 carry a cross-reference identifier, so a lookup route exists.

Table 4: Yeast9 species by structure class, with how many carry a SMILES.

structure_class	n_species	n_compartment_copies	n_with_smiles
specific_molecule	914	1742	858
acyl_resolved_lipid	258	692	0
trna	44	75	0
inorganic_or_ion	31	125	28
generic_r_group	29	42	5
protein_species	24	33	2
pooled_lipid_class	23	41	0
kegg_glycan	22	22	0
acp_thioester	16	16	0
pseudo_metabolite	11	12	0
no_formula_other	6	6	1

The consequence for anything reaction-level is that 2,156 of 4,131 reactions, 52 percent, have a structure for every participant.

The axes are related but not the same. Against the full five-dataset compound set, 49 exact keys and 103 connectivity skeletons are shared with Yeast9, so 103 metabolites are dosed as exogenous compounds somewhere in the five datasets. They are the central cofactors and metabolites rather than the drug-like library. The media axis is a different story: 30 of 37 media components are Yeast9 metabolites, which is expected since media components are metabolites by construction and is the strongest argument for one shared space. Because Yeast9 is stereo-flat while the curated compound keys carry stereochemistry, any join between the axes is on the connectivity skeleton with the ambiguity recorded.

7 The representation x

Figure 1 is the input representation the preceding sections argue for. It has three channels and one shared encoder, and it carries its own limits.

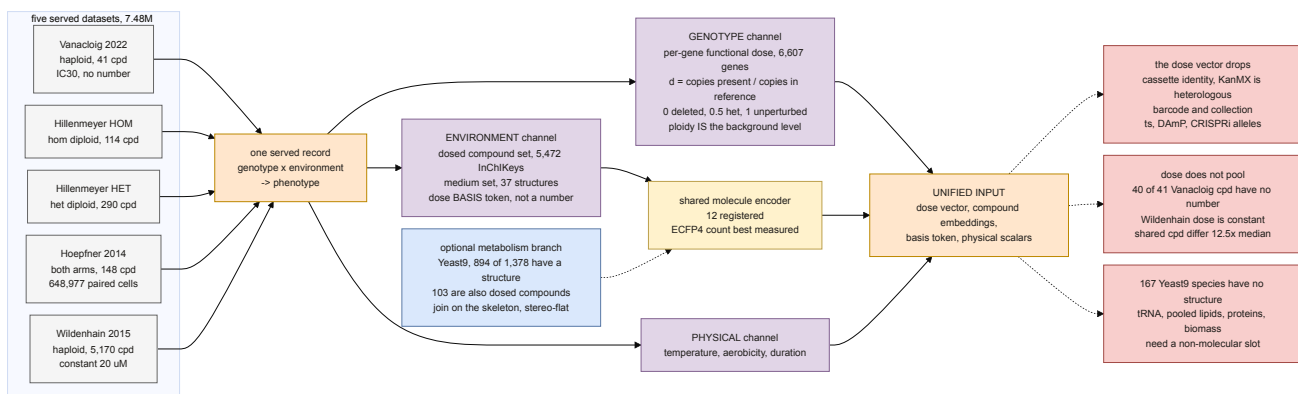


Figure 1: One input representation across the five datasets. Gray is a served dataset, purple a per-record input channel, yellow a fixed table, orange a computation, blue the optional metabolism branch, red a measured limit the representation does not carry. The genotype channel is a per-gene functional dose whose background level is the ploidy. The environment channel is a compound set through the shared molecule encoder plus a dose basis token rather than a concentration. The metabolism branch reuses the same encoder.

Genotype. A per-gene functional dose over the gene universe, copies present over copies in the reference. Zero for a deletion, one half for a heterozygous deletion in a diploid, one for an unperturbed gene. Ploidy needs no separate

feature because it is the background level. This is lossless for these five datasets and is validated by Hoepfner’s 648,977 paired cells.

Environment. The set of dosed compounds as InChIKeys through the shared molecule encoder, the medium components through the same encoder, and a dose basis token rather than a concentration. The token is the honest encoding given Section 4: a target-inhibition dose, a fixed molar dose and an unstated dose are three different statements, and a single number would assert a comparability that does not exist.

Physical. Temperature, aerobicity and duration as scalars.

What it does not carry, and why that is stated rather than hidden. The dose vector drops cassette identity, barcode and collection, and every allele class off the dosage axis. Dose does not pool. And 167 Yeast9 species have no structure and need a non-molecular slot. Each of those is a measured limit rather than an anticipated one, and each bounds a specific later step: the allele classes bound any join to the trigenic datasets, the dose limit bounds any cross-dataset dose-response claim, and the structureless species bound the metabolic branch.

Two decisions this leaves open. Whether to keep the Hillenmeyer homozygous arm, which contributes 1.09 million records and one non-exact chemical neighbor above 0.5, is a judgment about whether record count without chemistry is worth the pooling cost. And whether the functional dose should scale the existing cross-attention perturbation operator or enter as a separate feature is an architecture question this document does not settle; the operator is already a soft mechanism rather than a mask, so scaling it is the smaller change.